A stricter forecasting study

This study is separate from the approved buyer-facing replay. The protocol was written before fitting, candidates and rules were fixed, and no model can promote itself into the application.

The point-error result

The strongest observed candidate was weather individual · model weight 0.5, with a 6.60% pooled development basket MAE improvement over latest price. It did not meet the required 10% improvement. No learned candidate passed every guard, so the research selection remains the latest-price baseline.

The interval result

The selected relative joint-residual range covered 71.67% of development outcomes for a nominal 80% range. The additive range covered 67.67%. Neither reached the predefined 75% minimum coverage guard. The study is not eligible for promotion.

Chronological design

Validation	Train through	Calibration	Cal baskets	Validation baskets
2023Q1	2022-09-30	2022-10-01 to 2022-12-31	56	57
2023Q2	2022-12-31	2023-01-01 to 2023-03-31	57	46
2023Q3	2023-03-31	2023-04-01 to 2023-06-30	46	48
2023Q4	2023-06-30	2023-07-01 to 2023-09-30	48	56
2024Q1	2023-09-30	2023-10-01 to 2023-12-31	56	54
2024Q2	2023-12-31	2024-01-01 to 2024-03-31	54	39

Six quarterly validation windows cover 2023 Q1–2024 Q2. The preceding quarter is reserved for interval calibration; earlier 2022+ data train the model. Seven-day targets crossing any boundary are purged. Each fold requires at least 30 complete calibration baskets and 30 validation baskets; at least four usable folds are required.

Target: the average observed wholesale point price in issue+1 through issue+7, with at least four quotes per vegetable. Reference basket: 20 kg beans, 30 kg carrots and 20 kg tomatoes. Weather features end seven days before issue.

What was compared

Four baselines: latest price, 28-day observed mean, feasible prior-year weekly price, and an observed-price EWMA with a seven-calendar-day half-life. Four existing LightGBM price/weather pooled/individual families are evaluated at model weights 0.25, 0.50, 0.75 and 1.0, blended with the latest price. This produces 20 candidates. LightGBM complexity settings remain unchanged.

Candidate	Basket MAE · LKR	Crop guard	Improved folds
weather individual · model weight 0.5	2,398.23	Pass	5
price individual · model weight 0.5	2,410.89	Pass	4
price individual · model weight 0.75	2,431.55	Pass	2
weather individual · model weight 0.75	2,433.42	Fail	5
price pooled · model weight 0.5	2,436.13	Pass	3
price pooled · model weight 0.75	2,438.70	Pass	2
weather individual · model weight 0.25	2,448.38	Pass	5
price individual · model weight 0.25	2,457.95	Pass	4
price pooled · model weight 1	2,473.70	Fail	2
price pooled · model weight 0.25	2,475.00	Pass	5
weather pooled · model weight 0.5	2,479.25	Pass	4
weather pooled · model weight 0.25	2,491.47	Pass	4
weather pooled · model weight 0.75	2,511.22	Fail	4
price individual · model weight 1	2,524.52	Fail	2
weather individual · model weight 1	2,539.52	Fail	4
latest price	2,567.62	Baseline	—
weather pooled · model weight 1	2,626.41	Fail	4
ewma 7d	3,817.57	Baseline	—
mean 28d	4,992.70	Baseline	—
prior year	8,111.51	Baseline	—

All qualifying candidates are considered before selecting the lowest-error candidate. Qualification requires at least 10% lower pooled basket MAE than the strongest development baseline, no vegetable MAE regression above 5%, and improvement in at least four folds. The baseline is retained if none qualifies.

Show the weak spots

Selected development basket MAE: LKR 2,567.62 across 300 baskets. Its 95% block-bootstrap interval is LKR 1,788.29–3,517.51. Relative range coverage is 71.67%, with a 95% block interval of 63.09%–79.75%.

The best observed candidate's gain interval is -2.84% to 15.72%. It includes no improvement. Results are development selection evidence, not independent confirmation of a winner.

Segment	Baskets	Basket MAE · LKR	Coverage	Mean width · LKR
volatility low	87	1,827.76	78.16%	7,361.01
volatility medium	90	2,944.83	64.44%	6,979.12
volatility high	123	2,814.94	72.36%	8,289.59
quarter 2023Q1	57	1,452.89	73.68%	4,355.93
quarter 2023Q2	46	1,568.75	82.61%	6,169.75
quarter 2023Q3	48	1,446.44	68.75%	8,280.18
quarter 2023Q4	56	1,907.68	76.79%	5,712.48
quarter 2024Q1	54	6,737.18	44.44%	13,300.87
quarter 2024Q2	39	1,929.33	89.74%	8,216.85

Volatility regimes use training-only thresholds of a quantity- and latest-cost-weighted trailing 28-day coefficient of variation. No future prices define a regime. Intervals preserve aligned commodity residuals rather than adding marginal endpoints.

Bootstrap: 2,000 resamples, seed 42, shared 14-calendar-day blocks within folds (38 blocks). Overlapping weekly targets remain dependent; these intervals are not conditional-coverage guarantees. Interval methods are ranked by interval score subject to the development coverage guard.

Keep research separate

Previously inspected period	Baskets	Basket MAE · LKR	Coverage
2025	201	2,674.05	78.11%
early_2026	41	2,227.45	82.93%

The 2025 and early-2026 results were already inspected during previous project work. They are labelled exploratory historical checks. They cannot establish fresh generalization for this study. A new protocol and new uninspected outcomes are required before any model promotion.

Implementation and reproduction

Run python -m marketbudget.research. The immutable run directory contains protocol.json, results.json and development_predictions.csv. The current study is exposed in a separate research panel and through GET /api/v1/research/latest. Research runs never write the approved forecast pointer or published evaluation.

Current limitations

The source series ends on 10 March 2026. The primary-source sample audit is partial, especially 2022. Source publication times and historical weather vintages are unknown. CBSL upstream reuse terms remain unconfirmed. There is no completed buyer pilot, invoice-cost validation or verified savings claim.

Provenance

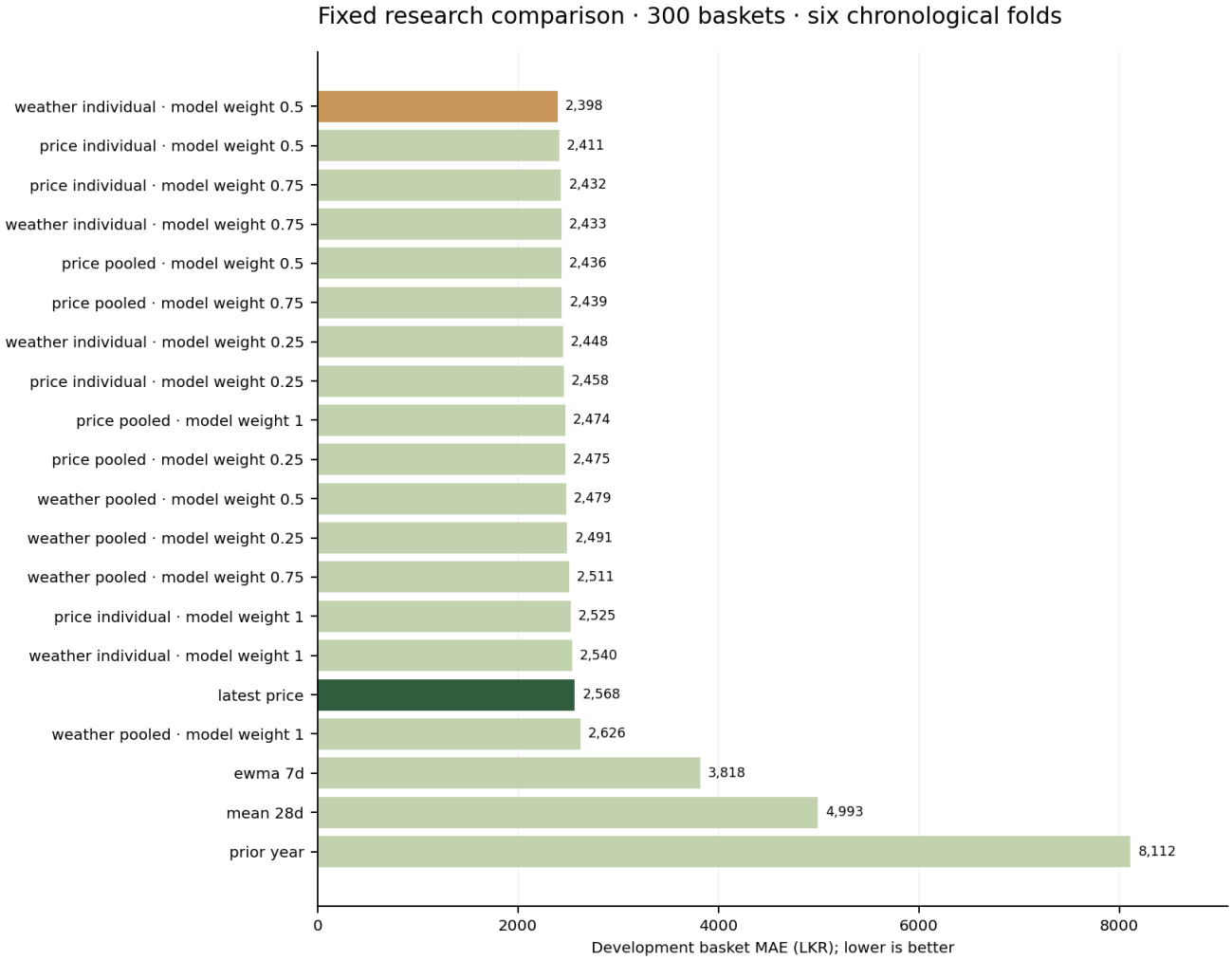
Study ID: study-a172c38095d77efd

Frozen protocol SHA-256: a172c38095d77efdeecdddcdd83aa4dfe60b01dcdeb4a241784802d90f5b2aca

Inputs: vegetablesSriLanka 1.1.0 CBSL-derived Dambulla wholesale observations, retained 2022+; NASA POWER regional proxies. Dataset hashes, code hashes, package versions, folds, parameters and selection rules are preserved in the protocol.

OpenAI Codex assisted in implementation, research execution and reporting. The original submission team should review and own the result. This report documents unsuccessful guards as well as promising comparisons.

All candidate errors



Green: selected baseline. Orange: lowest observed development error. No learned candidate cleared the full qualification rule. The displayed errors share the same complete-basket issue dates within each fold.